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Photoplethysmogram at green light: Where does chaos arise from?

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ABSTRACT

Photoplethysmography has been routinely used for health monitoring in the past decades. Even though in daily medical practice, photoplethysmograms (PPGs) are recorded at red and near infra-red light (rPPGs), during the last decade, PPGs obtained at green light (gPPGs) have been widely used in wearable devices, such as wristbands and smartwatches, thus providing highly usable and accessible daily health monitoring. It is well recognized that PPG signals obtained at NIR light contain sufficient information about a person's health; furthermore, they are well-studied and highly complex. By contrast, green light PPGs have not been sufficiently studied; thus, it is not clear whether the provided information is as valuable as that obtained by rPPGs, as green light photoplethysmography is formed at papillary dermis and higher skin levels. However, the gPPG signal is recognized to be more robust to motion artifacts compared with rPPGs. As rPPG dynamics has been recognized as chaotic, this study is aimed at investigating the properties of gPPGs and compare them with those of rPPGs to determine whether gPPG dynamics is chaotic as well. The motivation for this question was not only to investigate whether gPPGs can be used for the same range of applications as rPPGs but also to obtain insight into whether the complexity of rPPGs is solely due to processes in deeper layers of the tissue or it is influenced by the higher-skin layers processes that create gPPGs. The obtained results demonstrated that gPPGs, as well as rPPGs, are chaotic, which implies that processes in the upper layers create chaos in PPG data, whereas the contribution of arterial blood volume changes is not completely clear.

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1. Introduction

Since the introduction of photoplethysmography in 1937 by A. Hertzman [1], it has significantly been developed, and presently, it is widely utilized in various medical applications. Equipment based on this technique is routinely used for heart rate and oxygen saturation estimation in clinical settings as well as wearable devices. Photoplethysmography is an optical non-invasive measurement technique in which a source of light is used to illuminate the skin tissue, and based on the device design, transmitted or reflected light intensity is collected by a photodetector to record the photoplethysmogram (PPG). Recorded light intensity variations have been traditionally thought to be associated with blood volume pulsations in the microvascular bed of the tissue.

For decades, near-infrared (NIR) light has been used as a light source for tissue illumination. However, it has recently been reported that blue, green, and even ambient light can be also utilized for recording PPGs [2–8], and all these signals can be used for calculating heart rates. Further, the absorption coefficient of the erythrocytes in blue-green region is seven times higher than for the red light [9]. These studies also demonstrated that green light PPGs (gPPGs) are more resistant to motion artifacts compared with red and NIR PPGs (rPPGs) and may provide more reliable results for heart rate estimation. Several recent wearable devices, such as

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wristbands and smartwatches, have been utilizing green light for pulse monitoring [8].

Tissue penetration depth of green light (530 nm) is less than that of red (660 nm) and NIR light (900 nm) [10]. Thus, green light reaches the epidermal-dermal junction and papillary dermis, whereas red and NIR light can reach dermal, subcutaneous, and deeper layers. According to conventional views, the PPG waveform is mainly due to blood volume changes in the vessels. However, green light penetration depth is not sufficient to reach large blood vessels where volumes vary according to the blood circulation phases, and therefore cannot directly read the blood flow associated information. With the upper-layer thinnest capillaries that act as rigid tubes [11,12], in a recent study on imaging PPG obtained at green light [9,12,13], it was demonstrated that the PPG signal may be formed by a different process than it is conventionally believed. In [12], a new model explaining PPG formation was proposed, according to which the PPG is created by the pulsatile transmural pressure of the arteries on capillaries that changes tissue optical properties, rather than by blood volume variation, as has thus far been believed. Thus due to the stretching and compression of the inter-capillary tissue with the heart beating the distance between nearby capillaries can change. This creates changes in the capillaries density in papillary dermis and as a result variation in the optical properties of the tissue [9,12]. However, it can be assumed that rPPGs are influenced by more factors than gPPGs, as green light is less penetrative than NIR. This fact gave rise to a new question, namely, whether the gPPG signal is formed by the same process as conventional NIR PPG signals, and whether gPPGs have the same complexity as rPPG processes.

A complete understanding of the PPG process is still lacking [12–14]. Even though nearly one century of various NIR PPG studies support the conventional point of view according to which the blood volume primarily contributes to the PPG waveform and defines its dynamics, most recent results provide evidence that several different processes create PPGs. PPG devices are in daily use and have a potential for various applications in physiological and mental health monitoring [14–18]. Thus, understanding which process actually creates PPGs may be of critical importance.

In numerous studies devoted to PPGs, various aspects of the PPG signal itself have been investigated, such as PPG signal dynamics [19], as well as its possible applications [14,17,18,20–25] and analysis methods [26–29]. Most of these studies were concerned with NIR PPGs. Previously, it was shown that the rPPG signal is consistent with the definition of chaotic motion [15,19]. In contrast with conventional signal analysis methods, such as Fourier or contour analysis, that are often applied to PPGs, the analysis of the chaotic characteristics of rPPGs demonstrated properties indicating their complex nature and provided essential information regarding their signal dynamics. However, if gPPGs are influenced by fewer factors and are less complex than rPPGs, as it may be assumed, it is unclear whether the gPPG signal is also chaotic and whether methods of nonlinear time series analysis may prove meaningful and beneficial for applications. Furthermore, if the recently suggested new model for PPG signal formation is correct, then there is no reason to expect that gPPG chaotic properties are significantly different from rPPG chaotic properties. Moreover, the processes in the upper parts of the tissue can in fact be those that create chaos in the PPG data.

Taking into account these considerations, investigation of the chaotic characteristics of gPPGs in comparison with the same characteristics of rPPGs may provide insight into whether the gPPG signal is less complex than the rPPG signal, emphasize the importance of physiological processes occurring in upper layers of the tissue, and provide additional support to the newly suggested PPG formation model proposed in [12]. Thus, if there are considerable similarities between rPPG and gPPG characteristics, and the gPPG signal is chaotic, then there is strong evidence that both the rPPG and the gPPG signals are to a significant extent formed by the same process, and the PPG dynamics does not solely depend on the blood volume variations at the deeper levels of the skin tissue. This would also indicate that the PPG complexity arises from processes in the epidermis, epidermal-dermal junction and papillary dermis rather than at dermic and subcutaneous levels. However, if gPPGs exhibit less complex dynamics, then there is solid evidence that the light penetration depth into the tissue has a critical effect on the dynamics of the PPG signal, and therefore cutaneous blood circulation in the arteries located on the deeper levels of the skin tissue affects the rPPG signal and creates chaos in it.

Additionally, understanding the gPPG characteristics may contribute to applied studies because, as was previously shown, even though rPPGs are valuable for various applications (other than conventionally obtained heart rates), they are highly movement-sensitive, whereas gPPGs could be more robust owing to their stability to movement artifacts.

In this study, methods of nonlinear time series analysis were utilized for investigating the gPPG chaotic characteristics, and the results were compared with the corresponding rPPG indexes to find evidence on whether the gPPG signal is chaotic and possesses dynamical properties similar to those of the rPPG signal, or the underlying signal dynamics is different. A comprehensive exploration of the rPPG chaotic properties was previously undertaken in [19]; however, for comparison between gPPGs and rPPGs, certain results corresponding to rPPGs are presented in this study as well. The possible gPPG dynamics is reconstructed by the time-delay embedding technique, determinism is tested by the Wayland test, predictability is tested by deterministic nonlinear prediction to which the method of surrogate data is applied, trajectory divergence is estimated by the largest Lyapunov exponent, and the 0–1 test for chaos is applied.

2. Methods and materials

2.1. Data collection

Data were collected from 18 subjects from the Institute of Vegetable and Floriculture Science (NIVFS), NARO workers. The data collection and analysis were approved by the NIVFS ethical committee. Informed consent was provided by each participant prior to the data collection. The participants answered a questionnaire regarding their health condition. According to the survey, all participants at the time of data collection were healthy non-smokers, physically active (on a similar level), did not take any medication, and did not have any history of heart diseases. The participants were 21–49 years old.

The PPG signal was collected by a finger reflectance-type recorder equipped with a green LED (530 nm) for gPPG recording, and by a finger transmission-type recorder with a NIR light LED (900 nm) for rPPG recording. The recorders were of different types to differentiate the effect on the rPPG signal of processes in the dermis and subcutaneous layers and the effect on the gPPG signal of processes in layers up to the papillary dermis.

The gPPG signal was recorded by a PPG recorder based on an Arduino UNO board equipped with a PPG sensor for Arduino; the rPPG signal was recorded by a finger recorder (I.W. Technology Firm, Inc. Tokyo Devices). The data sampling frequency was 500 Hz for gPPG and 400 Hz for rPPG, respectively. The data collection was designed considering factors that may affect cutaneous blood flows [30,31]. The data recording was conducted in a sitting-relaxed position, in a quiet air-conditioned room at a temperature of 24 and 25 °C and moderate ambient light. Additionally, the participants were asked to refrain from movements of the arm on which the signal was recorded. The data collection experiment consisted of

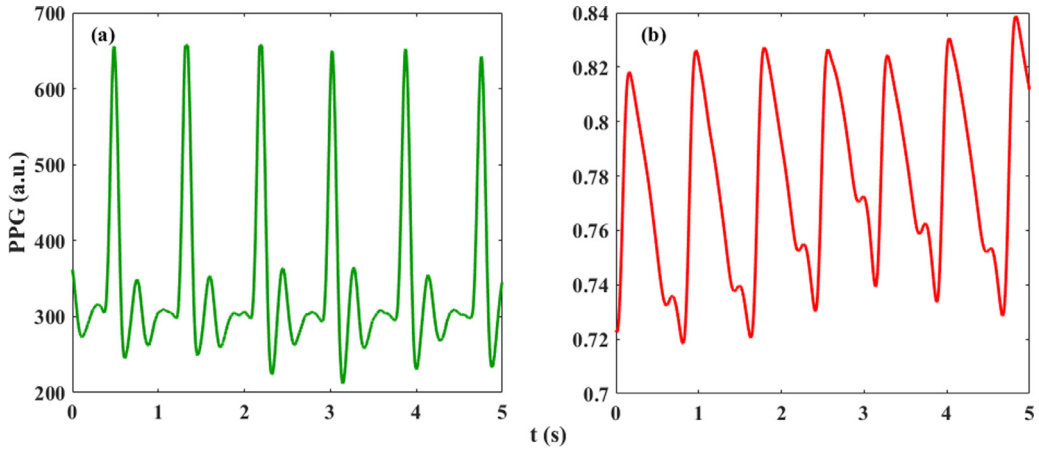


Fig. 1. 5-s long portion of the signal from healthy subjects (a) gPPG and (b) rPPG.

two parts. In one part, gPPG and rPPG data were collected simultaneously from eight subjects; the gPPG recorder was applied to the right's hand forefinger, and the rPPG recorder was applied to the left hand. In the other part, additional gPPG data were collected from another ten subjects. In both experiments, for each subject, two measurements were repeatedly performed, and each measurement was 5 min long. It is important to notice that data collected from the left and right hand may have some differences owing to the measurement site; however, these differences are insignificant if the data are recorded from a healthy subject [31].

The participants were questioned about the level of comfort and temperature sensation at the experiment site based on the scale proposed in [32], consisting of four levels of comfort sensation (comfortable, slightly uncomfortable, uncomfortable, and very uncomfortable) and seven levels of thermal sensation (cold, cool, slightly cool, neutral, slightly warm, warm, and hot). According to the answers, the experiment environment was estimated as comfortable by 78% participants and slightly uncomfortable by 22%; temperature sensation was estimated as neutral by 44%, slightly cool by 39% and cool by 17% of the participants. Data collection was conducted in a summer season. Typical examples of a 5-s long portion of gPPGs and rPPGs obtained from two different subjects aged below 40 years are shown in Fig. 1.

3. Results

3.1. Reconstruction of possible dynamics

Most methods of nonlinear time series analysis are based on time-delay reconstruction. This allows obtaining possible dynamics based on a single variable, which is in several cases the only known information obtained during an experiment. If x_1, x_2,..., x_n is the investigated time series, its dynamics can be reconstructed in m-dimensional phase-space by the time-delay embedding method with time-lag τ. Then, points on the reconstructed trajectory in time-delay coordinates can be defined as $X_i=(x_i, x_{i+\tau}, \ldots, x_{i+\tau(m-1)})$. The dimension m for reconstruction was chosen based on the false nearest neighbors method [33], as shown in Fig. 2. As seen in Fig. 2. $m=4$ and 5 can be an appropriate choice for the embedding dimension. However, between $m=4$ and $m=5$ the differences of the quantitative results of further discussed methods were minor. Therefore, all the results shown in this paper correspond to the smallest embedding dimension $m=4$. Time-delay reconstructed rPPG dynamics had previously found application in mental health monitoring [15,16]. Measured gPPG data were reconstructed by the time-delay embedding technique with time-lag $\tau=40$. Fig. 3 shows the time-delay reconstructed gPPG and rPPG trajectories. As seen in Fig. 3(a) and (b), the possible gPPG and rPPG dynamics trajectories reconstructed in the phase-space differ considerably.

Table 1
Values of the Wayland-test translation error (WTE), the largest Lyapunov exponent (LLE), and 0–1 test for chaos corresponding to the first measurement for each subject in simultaneously recorded gPPG and rPPG signals.

Subject		gPPG			NIR PPG		
No.	Age	WTE	LLE	0–1 test	WTE	LLE	0–1 test
1	25	0.023	1.195	0.995	0.053	0.834	0.996
2	28	0.074	0.615	0.995	0.011	0.708	0.994
3	36	0.079	0.895	0.990	0.026	0.985	0.979
4	39	0.050	1.679	0.997	0.037	0.852	0.993
5	42	0.092	0.628	0.996	0.024	0.786	0.993
6	42	0.100	0.285	0.993	0.040	0.446	0.992
7	46	0.066	0.864	0.996	0.032	1.132	0.994
8	49	0.050	1.340	0.994	0.018	0.629	0.993

3.2. Determinism estimation by the Wayland test

Determinism is an essential property of deterministic chaos. The Wayland test [34] provides quantitative information regarding determinism of time-series. The Wayland-test translation error was successfully used for determinism check in various experimental data [35–38], including PPG data [19,39]; moreover, it was found to be a valuable index for applied studies on rPPGs [18].

The Wayland test can be performed as follows. If $X_i=(x_i, x_{i+\tau}, \ldots, x_{i+\tau(m-1)})$ are points on the time-delay reconstructed trajectory of a time series x_1, x_2, ..., x_n, X_0 is a fixed and arbitrarily chosen point on the reconstructed trajectory, X_i $(i=1,2,\ldots,k)$ are its k nearest neighbors, which are contained in a sufficiently small vicinity of X_0, and y_i are images of X_i $(i=0,1,\ldots,k)$, then $v_i=y_i-X_i$ will be translation vectors. The local translation error (LTE) is defined by the following formula [34]:

$$e_{loc}=\frac{1}{k+1}\sum_{i=0}^{k}\frac{\|v_i-\langle v\rangle\|^2}{\|\langle v\rangle\|^2},$$

where $\langle v\rangle$ is the average of v_i. For the deterministic time series x_j $(j=1,2,\ldots,n)$, the translation vectors v_i will be nearly identical; therefore, the LTE will be equal or nearly equal to zero.

The Wayland-test translation error was calculated for all collected time series. For the gPPG data, the average Wayland-test translation error (WTE) was WTE = 0.069 and the average value for the rPPG data was WTE = 0. 073. The WTE values for the first measurement for simultaneously recorded gPPGs and rPPG are shown in Table 1. As was previously shown, gPPGs [39] and rPPGs

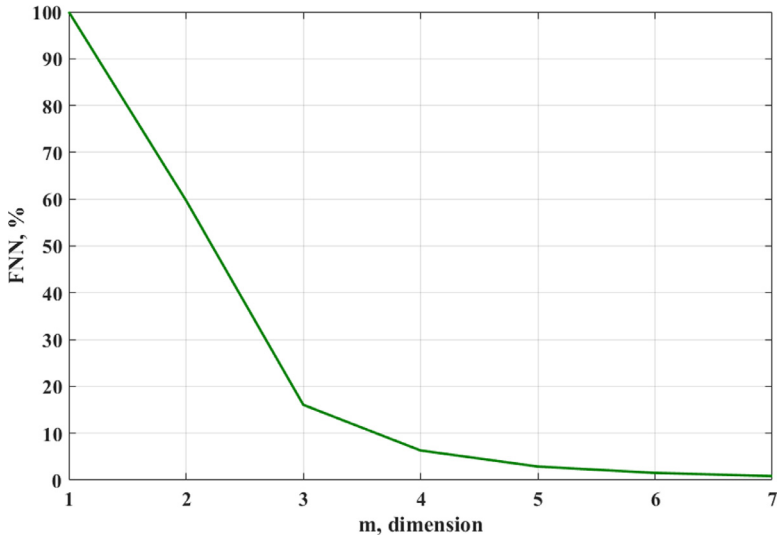


Fig 2. Estimation of the gPPG minimum embedding dimension by the false nearest neighbors (FNN) method.

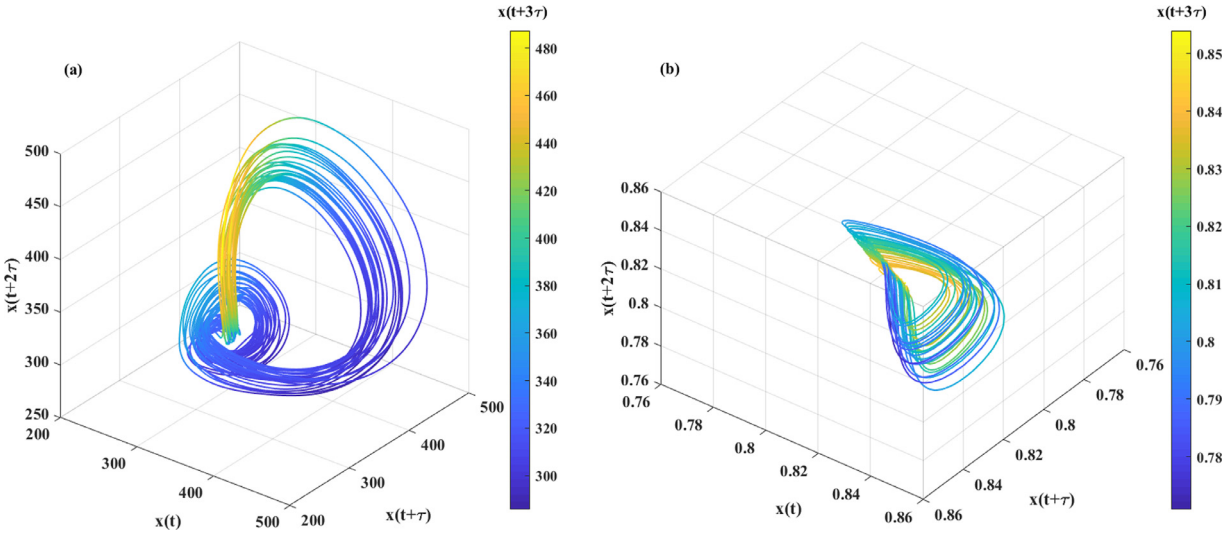


Fig. 3. Typical example of time-delay reconstructed trajectory with time-lag $\tau = 40$ for (a) gPPG and (b) rPPG.

[40] exhibit local noise sensitivity, i.e., the response of the trajectory to noise induction is not uniformly distributed along the reconstructed trajectory, and some regions of the trajectory may be affected by noise more than others. Fig. 4 shows the distribution of LTEs along the 10-s long portion of the gPPG signal. It can be seen that only certain regions on the time series have higher LTE values, whereas along most of the other parts of the trajectory, the LTE values are close to zero. Therefore, the WTE values obtained for the gPPG data can be recognized as sufficiently small, and this clearly indicates determinism.

3.3. Predictability

Most of the results of the Wayland test clearly indicate determinism of the data under investigation; however, the possibility of flaws in Wayland-test results was previously pointed out for the case of noise contaminated data, when the amount of noise is significant [41]. The investigation of data predictability is often an important part of chaos identification, as it can reveal such essential properties of chaotic dynamics as dependence on initial conditions and determinism. Previous studies on rPPG predictability demonstrated that long-term forecasting is impossible; however, short-term predictability exists [19,42]. This is one of the defining characteristics of chaotic time series. The existence of short-term predictability is also an indicator of determinism for data; moreover, it has important practical values. gPPG predictability was tested by deterministic nonlinear prediction (DNP). Considering noise-related issues in the experimental data and the problem of checking whether data are deterministic or the realization of a random process, the method of surrogate data was applied in addition to DNP. 50 surrogate data sets were generated for each time series by phase randomization of the original data's Fourier transform [43,44] to test the null hypothesis that the signal is the realization of a linear Gaussian stochastic process, and by pseudoperiodic surrogate data method to test the null hypothesis that observed signal is consistent with uncorrelated noise-driven periodic orbit [45]. The choice of these particular surrogate data methods was based on consideration with the PPG properties. Thus one of the most distinctive properties of the PPG is that its predominant frequency component in the Fourier spectrum corresponds to the heart rate. Even if nondeterminism of the PPG signal is assumed, still its power spectrum contains information on the heart rate. The Fourier transform phase randomized surrogate method preserves original signal's power spectrum in generated stochastic sur-

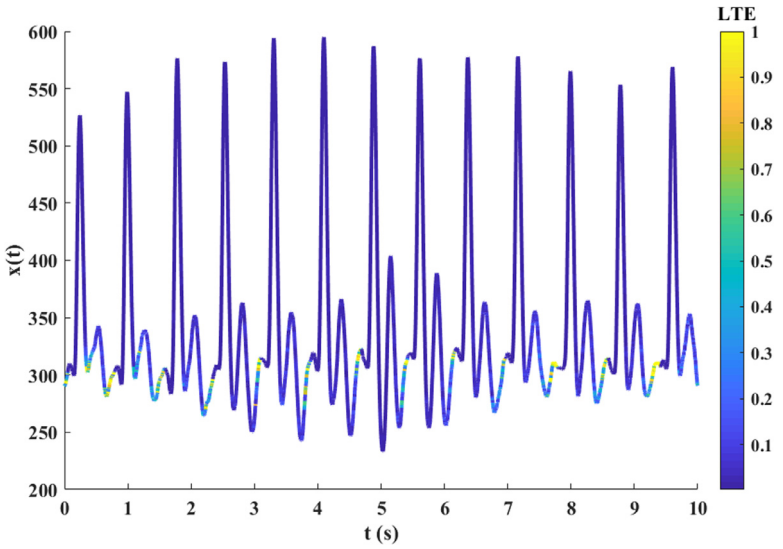


Fig. 4. Distribution of LTEs along the gPPG trajectory for a typical representative of gPPG time-series.

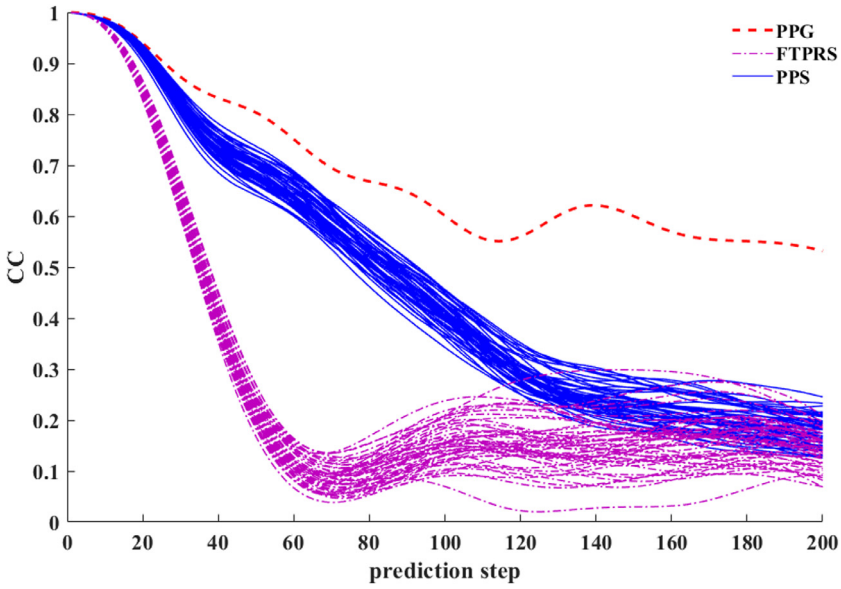


Fig. 5. Typical example of the CC between original and predicted time series for the gPPG (the red dashed line) and its 50 datasets of Fourier transform phase randomized (the purple dash-dotted lines) and pseudoperiodic (the blue lines) surrogates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rogate time series, therefore it was applied to the DNP results. At the same time, due to the nearly cyclic behavior observed in the PPG, for the several decades, there was a long-lasting debate on whether the PPG is genuine chaotic or it is a periodic signal with the signs of chaos due to the noise in experimental data. The DNP for noise-driven periodic orbits may often produce results similar to the case of chaotic time series. Paying attention to this issue the surrogate test for pseudoperiodic time series data was additionally employed. This method generates surrogate time series with a structure that matches an original data structure in a large scale and tests the null-hypothesis that the time series is an uncorrelated noise-driven periodic orbit [44,45].

Fig. 5 shows typical curves of the correlation coefficient (CC) between the original and the 200-step predicted time series for the original data and surrogate data sets. Fig. 6 demonstrates frequency distribution of the correlation coefficient for the 1- and 20-steps forward prediction. As seen in Fig. 5 and Fig. 6 prediction results corresponding to the original time series can be clearly distinguished from those for the both types of the surrogate time series, thus allowing the rejection of the null hypotheses under which the surrogate data sets were generated from the original data.

3.4. Trajectory divergence

The Lyapunov exponent, which quantifies the exponential divergence of two trajectories that are initially close to each other, is recognized as a hallmark of chaos and is commonly applied for chaos identification as well as for quantification of system properties observed under different conditions in various applied studies [26,46]. The largest Lyapunov exponent (LLE) was estimated in this study to quantify the differences between gPPG and rPPG data. The LLE was calculated by the method described in [47], whereby the average exponential separation of nearby trajectories is measured, i.e., $y(i) = \langle \ln d_j(i) \rangle / i\Delta t$, where $d_j(i)$ is the distance between the jth part of the nearest neighbors after time $i\Delta t$. The implemented

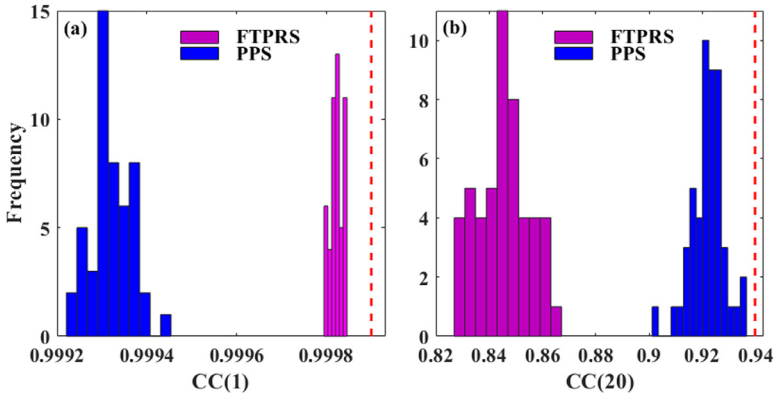


Fig. 6. Frequency distribution of correlation coefficient for 50 surrogate datasets of Fourier transform phase randomized (the purple bars) and pseudoperiodic (the blue bars) surrogates, and the corresponding correlation coefficient for original PPG signal (the red dashed line): (a) for 1 and (b) for 20 steps forward prediction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

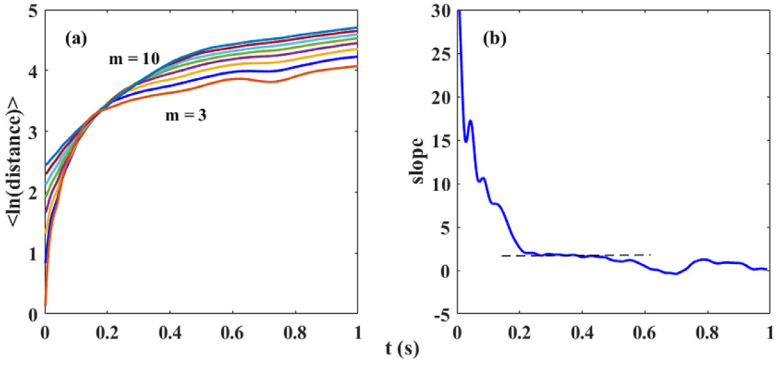


Fig. 7. LLE estimation: (a) Typical plot of the average exponential divergence versus time for embedding dimension $m = 3,\ldots, 10$; (b) slope corresponding to $m = 4$.

method is robust to the variation of the embedding dimension, the time-delay lag, and the noise level. It is worth noticing that unlike quasiperiodic or stochastic systems, which may be mistaken for chaotic systems, finite-dimensional chaotic systems exhibit convergence of the results once the dimension is sufficiently large. Fig. 7(a) shows the curves of the averages over all points of the logarithm of the divergence between neighboring trajectories, that is, $d(i) = \langle \ln d_j(i) \rangle$ calculated for dimension $m = 3,..,10$. The LLE is defined as the slope of a least-square linear fit to the $d(i)$ curve. The slope corresponding to dimension $m = 4$ is shown in Fig. 7(b). Examples of LLEs for the gPPG data and the simultaneously collected rPPG data are shown in Table 1; only the results for the first measurement are shown.

Numerous studies have pointed out that the presence of noise in the data may cause miscalculation of the LLE value [44,47]. As gPPGs and rPPGs are noise-contaminated experimental data, the obtained LLE values cannot be directly and unambiguously compared. However, as can be seen from the results of the Wayland test calculation, the amount of noise in the data is expected to be low; thus, the method implemented for the LLE calculation is expected to be sufficiently robust. As seen in Table 1, all LLE values corresponding to the gPPG and the rPPG data are positive, and no clear difference between them (independent of the subject) can be found.

3.5. 0 – 1 test for chaos

Most approaches for chaos identification in experimental data are based on the reconstruction of possible dynamics in the phase space. The "0–1 test for chaos" [48–53] is a test for chaotic behavior of time series that, unlike other methods, does not rely on phase-space reconstruction and is applied only to the time series itself. Owing to its simplicity and advantages, it is more attractive and reliable compared with, for example, the LLE index for chaos identification [53]. This binary test returns 0 for a regular dynamics system and 1 for a chaotic system. In the case of experimental data, a value close to 0 is treated as non-chaotic and close to 1 as chaotic [53].

For discrete time series, the 0–1 test for chaos can be performed as follows. Let $\{x_n\}$ $n = 1,\ldots,N$ be a discrete time series and $c \in R$ (in the present case, $c \in (0, 2\pi)$) a randomly chosen constant. For each c, the translation variables p_c and q_c are obtained as follows:

$$p_c(n) = \sum_{j=1}^{n} x(j)\cos(jc),\ q_c(n) = \sum_{j=1}^{n} x(j)\sin(jc),\ n = 1, 2, \ldots, N.$$

To analyze the behavior of p_c and q_c, their average mean square displacement is calculated by

$$M_c = \frac{1}{N}\sum_{j=1}^{N}\left((p(j+n) - p(j))^2 + (q(j+n) - q(j))^2\right),\ n \le \frac{N}{10},$$

with the growth rate

$$K_c = \lim_{n\to\infty} \frac{\log M_c}{\log n}.$$

Then, the value $K = median(K_c)$ takes either the value $K \approx 0$ for regular dynamics or $K \approx 1$ for chaotic dynamics.

The 0–1 test for chaos can be applied only to deterministic time series. The results of the Wayland test as well as the results of DNP and the surrogate data method indicated that the data under investigation are noise-contaminated deterministic time series, thus rendering the 0–1 test for chaos applicable to the gPPG and rPPG

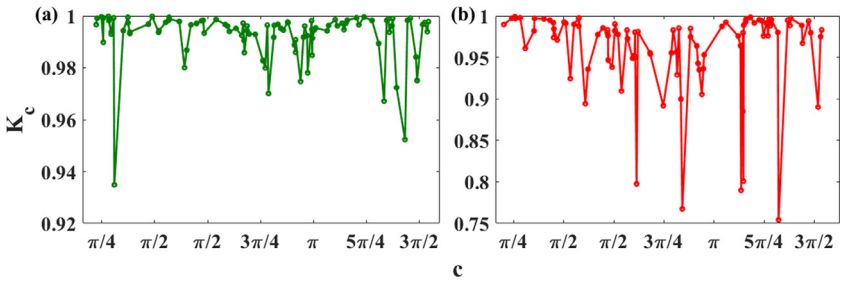


Fig. 8. Plot of K_c versus c for (a) gPPG and (b) rPPG.

data. The method is sensitive to oversampling, and the gPPG and rPPG signals were recorded at high resolution; therefore, the original time series were thinned to a 40 Hz sampling rate. Fig. 8(a) and (b) show a typical plot of K_c versus c for the gPPG and rPPG signals, respectively. The results of the 0–1 test for chaos for the first round of the simultaneously recorded gPPGs and rPPGs from the same subject are shown in Table 1. The average value obtained for the gPPGs was 0.996, and 0.993 for the rPPGs. These results indicate chaos in the studied data.

4. Discussion

The main objective of this study was to investigate whether the gPPG signal is chaotic and to study its dynamical characteristics. The rPPG chaotic characteristics were previously comprehensively studied and were found to be useful for various health monitoring applications; however, gPPG dynamics may be different from rPPG dynamics, and it has not yet been investigated whether it is chaotic and similar to rPPG dynamics. Through the comparison between gPPGs and rPPGs, it was studied whether the processes in the epidermis, epidermal-dermal junction and papillary dermis layers of the skin are responsible for complex chaotic PPG dynamics or the chaos observed previously in rPPGs is solely due to blood volume variations at deeper levels of the skin, such as dermis and subcutaneous layers, that cannot be reached by green light.

To test the determinism of the gPPG signal, the Wayland-test translation error was first calculated. As seen in Table 1, the WTE values are sufficiently small taking into consideration that experimental data inevitably contain a certain amount of observational noise. Additionally, as seen in Fig. 4, the distribution of LTEs is not even, and only small parts of the gPPG trajectory exhibit higher LTE values. These results are strong evidence of determinism in the gPPG data. However, taking into account possible flaws in determinism estimation in experimental signals contaminated with noise of unknown nature, making a conclusion based on a single test may lead to misidentification of determinism in the studied data. Therefore, additional effort was made to confirm determinism by conducting DNP combined with the Fourier phase randomized surrogate data method. Short-term predictability was found for the gPPG data, as well as significant differences in the DNP results for the original data and the surrogate datasets, which were obtained under the null hypothesis that the studied data are the realization of a linear Gaussian process, providing additional evidence on the deterministic nature of the gPPG data. Once the determinism of the data is confirmed, the 0–1 test for chaos can be conducted. This test performs well and is applicable strictly to deterministic systems. It returns 0 for regular and 1 for chaotic dynamics. As the gPPG data are noise-contaminated deterministic signals, and the values of the 0–1 test for chaos are sufficiently close to 1, it may be concluded that the gPPG signal is chaotic, as is the case with the rPPG signal.

Both gPPG and rPPG signals appeared to be chaotic even though they were obtained by illumination by light of different wavelength, and the gPPG system operates in reflectance mode, whereas a transmission-type rPPG recorder was used. Thus, taking into consideration the differences under which gPPG and rPPG signals were recorded and that both signals are chaotic, it may be concluded that processes occurring under the depth of green light penetration into the skin create the chaotic motion of gPPGs. However, unlike gPPGs, which are influenced by the changes in capillaries density at the papillary dermis, rPPGs are influenced by processes in deeper layers of the skin, such as blood volume variation, that were for a long time believed to be responsible for the PPG waveform. Therefore, it is not clear whether the chaos in rPPGs is solely due to the same processes that create gPPGs, and the blood volume variation of the arteries only contributes to the amplitude of the rPPG signal (and in fact “simplifies” it compared with the gPPG signal), or this process is chaotic as well. To obtain insight into this question, the chaotic characteristics of gPPGs and rPPGs were compared.

The properties of gPPGs that were chosen for the investigation were trajectory divergence (dependence on initial conditions) and predictability. To test the dependence on initial conditions, the LLE was calculated for all time series. As seen in Table 1, all the LLEs are positive, as is the case with chaotic time series, and there are no significant observable subject-independent differences in the LLE values between the gPPG and the rPPG signals. The average gPPG LLE value is 0.938 and the corresponding rPPG value is 0.797. Although several studies [44] indicated that most numerical algorithms for Lyapunov exponent estimation for discrete time series with noise can overestimate the LLE, considering the results of the Wayland test, it can be assumed that the amount of noise in the data is not sufficiently large to significantly alter the LLE values.

Predictability was estimated by DNP. Fig. 5 shows a typical example of the CC between predicted and actual signals for a 200-step forward prediction. The method of pseudoperiodic surrogates was additionally applied to the DNP to address the issue of possible similarities of the DNP results between chaotic and noise-driven periodic time series. Even though for both the gPPG (Fig. 5) and the rPPG signals [19,42], short-term prediction is possible, as can be seen in Fig. 5, the first CC curve of the gPPG signal decreases faster compared with previously reported results for rPPGs, i.e., the quality of prediction for gPPGs decays faster. This result does not contradict the difference in the LLE values for gPPGs and rPPGs, as the gPPG value, which is in average larger, indicates faster divergence of the neighboring trajectories. It is important to notice here that despite the difference in the data collection method, both types of signals have similar predominant frequency component corresponding to the subject’s heart rate.

Based on the obtained results, it can be concluded that although both signals are chaotic, gPPG chaotic characteristics seem to be more emphasized compared with rPPG characteristics. It can be suggested that the processes that contribute to the formation of the rPPG signal, other than that “creates” the gPPG waveform, do not increase the complexity of the rPPG data. Nevertheless, it is clear from the obtained results that the gPPG signal is chaotic, and

the processes at the upper levels of the skin are responsible for the gPPG complexity and, most likely, contribute to the formation of the rPPG signal.

Additionally, comparison of the related LTE distributions along the time series, which locally characterizes the determinism, may provide a partial explanation of the motion artifact "robustness" of gPPGs. Therefore, the obtained results not only provide a reply to the theoretical side of the question of whether the gPPG signal is consistent with chaotic motion but also are useful for health monitoring applications by providing insight into the pertinent gPPG chaotic properties.

5. Conclusion

The gPPG dynamics was investigated in order to answer the question whether it is chaotic and thus whether the rPPG chaotic dynamics is created in the upper layers of the skin, owing to changes in the capillaries density by the arterial transmural pressure, or in deeper layers, owing to changes in the vessel blood volume. Moreover, gPPG and rPPG chaotic characteristics were compared to clarify whether the gPPG signal might be suitable for applications in which the rPPG signal was previously proven useful and promising.

The results demonstrated that gPPG dynamics is chaotic, and its properties indicated the greater complexity of the gPPG signal compared with the rPPG signal; thus, the dependence on initial conditions is more emphasized in gPPGs. Although it is difficult to draw an unambiguous conclusion on whether the rPPG chaotic dynamics is solely due to processes in layers higher than dermis and blood volume changes have regular dynamics that only contribute to the PPG signal amplitude, or the rPPG signal is created by the interaction of several complex processes, there is no doubt that processes in the higher skin layers are those that create additional chaotic dynamics of gPPGs.

In addition to the understanding of the fact that the gPPG signal is chaotic, it can also be considered that gPPGs can be valuable not only for heart rate estimation, as they have thus far been used, but also, using nonlinear time series analysis, for rPPG applications such as mental and physiological health monitoring. These applications involve analysis of chaotic characteristics that are similar to those of gPPGs.

The obtained results improve the understanding of the PPG signal dynamics as well as its waveform creation process in general and the gPPG signal in particular. Moreover, they emphasize that the gPPG signal can be promising in applications of the rPPG signal, as their properties exhibit similarities.

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Author Contributions

N.S. conceived the idea and the experiment. N.S. and T.Z. designed and prepared the experiment settings. T.Z. conducted the data collection and performed the data preprocessing. N.S. analyzed the data and wrote the first draft of the manuscript. K. A. revised the manuscript. All authors participated in the results discussions and contributed to writing the final manuscript. All authors have read and approved the final manuscript.

Conflicts of Interest

The authors declare no conflict of interest.

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